

How We Researched This Task — Google, AI & Investigation Techniques

Professional Engineering Reflection

Database Management Systems

Task Type: Full-Day Investigation & Reflection

Date: 2026

□ How We Researched This Task — Google, AI & Investigation Techniques

Purpose of this file: This document explains, in full detail, **how we researched and built** the Enterprise Database Investigation report. We did not rely on AI alone — we combined **Google search techniques, official documentation, industry sources, and AI-assisted reasoning** into a complete research methodology. This file shows every technique we used, the exact questions we asked, and how we verified everything.

1. Our Research Philosophy

The task is a **professional investigation** — not a copy-paste exercise. Real software engineers research the same way:

Search → Read → Compare → Verify → Reflect → Write

We treated the investigation as if we were a **junior architect doing a real technology selection** for a company. That meant using **multiple research channels in parallel**:

Channel	Used for
□ Google Search	Finding articles, blog posts, comparisons, case studies
□ Official Documentation	Verifying facts about each DBMS
□ DB-Engines & Analyst Reports	Popularity rankings and market data
□ YouTube	Visual explanations of architecture (CDB/PDB, RAC)
□ Stack Overflow / Reddit	Real engineers' opinions
□ AI Assistant	Brainstorming, structuring, comparing, simplifying

The combination of all six was much more powerful than any single one alone.

2. The Full Research Workflow We Followed

We followed this exact step-by-step process for every part of the task:

```
STEP 1 – Read the task carefully
STEP 2 – Break it into small focused questions
STEP 3 – Google search with smart keywords
STEP 4 – Read official docs to verify
STEP 5 – Ask AI to compare, simplify, and structure
STEP 6 – Cross-check AI answers with sources
STEP 7 – Rewrite in our own words + add reflection
STEP 8 – Organize into the final document
```

PART A — HOW WE SEARCHED IN GOOGLE

This was our **primary research channel**. Google was used to find facts, articles, case studies, and official documentation.

3. The Google Search Techniques We Used

3.1 Keyword Search (Not Sentences)

We never typed full sentences into Google. We typed **3-6 keywords** that targeted the answer directly.

Bad search	Good search
"Can you tell me about Oracle database editions?"	oracle xe vs se vs ee comparison
"Why do banks use Oracle?"	why banks use oracle database
"I want to know what CDB and PDB are"	oracle multitenant CDB PDB architecture

3.2 Search Operators (Advanced Google)

We used Google's hidden operators to filter results precisely:

Operator	What it does	Example we used
" "	Exact phrase match	"pluggable database"
site:	Search inside one site	oracle RAC site:oracle.com
filetype:	Find PDFs or specific files	database comparison filetype:pdf
-	Exclude a word	mysql tutorial -beginner
OR	Either word	postgresql OR mysql cloud
intitle:	Word must be in title	intitle:"oracle vs postgresql"
2024..2026	Date range	database trends 2024..2026

3.3 The Exact Google Queries We Ran

These are the **real searches** we performed during the investigation, grouped by task part:

For Part 1 (4 DBMSs):

```
oracle database use cases enterprise
companies using oracle database list
sql server vs oracle comparison 2025
postgresql strengths and weaknesses
mysql vs postgresql startup
"why companies use sql server" site:microsoft.com
database licensing cost comparison
top open source databases 2025
```

For Part 2 (Oracle deep dive):

```
oracle xe limitations 21c
oracle standard edition vs enterprise edition features
oracle multitenant architecture explained
"container database" "pluggable database" tutorial
oracle PL/SQL vs SQL difference
oracle RAC architecture site:oracle.com
oracle data guard explained
why banks use oracle database
```

For Part 3 (Decision thinking):

```
why banks prefer commercial database
postgresql success stories startups
microsoft sql server market share
cloud database vs on-premise enterprise
total cost of ownership database
vendor lock-in database risks
```

For Part 4 (Architecture):

```

database scalability comparison
database high availability comparison
oracle RAC vs sql server alwayson
postgresql replication vs mysql replication
TPC-C benchmark databases
RMAN backup strategy oracle

```

For Part 5 (Migration):

```

amazon migration from oracle to aurora
uber postgresql to mysql migration why
database migration risks enterprise
oracle to postgresql migration tools
ora2pg tutorial
why migration projects fail

```

3.4 How We Picked Which Search Result to Read

Not every Google result is trustworthy. We used this **3-tier priority** when scanning the results page:

Tier	Source type	Trust level
□ Tier 1	Official docs (oracle.com, microsoft.com, postgresql.org)	Highest — used as source of truth
□ Tier 2	Analyst sites (Gartner, DB-Engines, AWS Blog, EDB)	High — used for trends and comparisons
□ Tier 3	Stack Overflow, Reddit, Medium, GeeksforGeeks	Medium — used for opinions and examples
△ Skip	Random SEO blogs, ads, content farms	Avoided — unreliable

3.5 How We Read Each Article (Speed Reading Technique)

Instead of reading every word, we used the "**Skim → Spot → Capture**" method:

1. **Skim** the headings to see the structure
2. **Spot** the key sentences (usually first sentence of each paragraph)
3. **Capture** the 2–3 facts we needed into our notes file

This let us process **10–15 articles per hour** instead of 2–3.

3.6 Multi-Engine Search

We didn't only use Google. We also used:

- **Bing** — sometimes shows different enterprise/Microsoft results
- **DuckDuckGo** — no tracking, more neutral results
- **Google Scholar** — for academic papers on database architecture
- **YouTube** — for visual architecture explanations (search: "Oracle CDB PDB explained")

PART B — HOW WE USED AI

AI was used as a **research partner and reasoning assistant** — not as the source of truth.

4. Why We Used AI Alongside Google

What Google does well	What AI does well
Finds primary sources	Explains complex concepts simply
Confirms facts and numbers	Compares many options at once
Surfaces real case studies	Structures answers into tables
Provides recent news	Generates reasoning paragraphs
Shows multiple perspectives	Simulates expert roles (architect, DBA)

Using both together gave us **breadth from Google + depth from AI**.

5. The 4 AI Prompting Techniques We Used

Technique 1 — The Role Prompt

We told the AI **who to act as** so it would respond like an expert.

□ Example prompt: "Act as a senior database architect with 15 years of enterprise experience. Explain the difference between Oracle Standard Edition and Enterprise Edition, focusing on what enterprises actually care about."

Why it works: AI adjusts depth and vocabulary based on the role. A "senior architect" gives more strategic, business-aware answers than a generic explanation.

Technique 2 — The Structured Output Prompt

We asked AI to return information in a **specific format** so we could reuse it directly.

□ Example prompt: "Compare Oracle, SQL Server, PostgreSQL, and MySQL across these 8 dimensions: scalability, performance, security, licensing, support, cloud readiness, backup, integration. Return the answer as a markdown table."

Why it works: Tables and structured answers are easier to verify, edit, and paste into the final report.

Technique 3 — The Reasoning Prompt

For Part 3 (decision thinking), we asked AI to **explain reasoning**, not just facts.

□ Example prompt: "Explain step-by-step why a bank would prefer Oracle over MySQL. Give 3 concrete technical reasons and 2 business reasons, with real-world examples."

Why it works: Forcing the AI to structure reasoning produces much higher-quality reflection content.

Technique 4 — The Devil's Advocate Prompt

After getting an answer, we asked AI to **challenge itself**.

□ Example prompt: "Now play devil's advocate. Why might a bank actually choose PostgreSQL or a cloud-native database instead of Oracle? Give counter-arguments."

Why it works: Exposes weaknesses in the original answer and produces a balanced, mature reflection.

6. The Exact AI Prompts We Used (Task by Task)

□ For Part 1 — DBMS Investigation

1. "List the top 5 industries that primarily use Oracle Database, and explain why each industry chose it."
2. "What are the 3 biggest strengths and 3 biggest weaknesses of Microsoft SQL Server in enterprise environments?"
3. "Why is PostgreSQL becoming so popular in cloud-native and startup environments? Give technical and business reasons."
4. "Compare MySQL vs PostgreSQL for a SaaS startup with 1 million users. Which would you recommend and why?"

5. "What is the typical licensing model and approximate cost for each of these databases: Oracle, SQL Server, PostgreSQL, MySQL?"
6. "Which official tools are used to manage each of these four DBMSs? Mention the IDE, backup tool, and monitoring tool for each."
7. "How does each of these databases integrate with AWS, Azure, and Google Cloud?"

□ For Part 2 — Oracle Deep Dive

1. "Explain the differences between Oracle XE, Standard Edition 2, and Enterprise Edition. Include limits on RAM, CPU, features, and price ranges."
2. "Explain Oracle's multitenant architecture (CDB and PDB) in simple terms with an analogy a junior developer would understand."
3. "Draw a text-based diagram showing how a CDB holds multiple PDBs."
4. "What is PL/SQL? How is it different from regular SQL, and why is it important in Oracle enterprise systems?"
5. "Explain Oracle RAC, Data Guard, GoldenGate, ASM, and Exadata in one paragraph each. Focus on what business problem each solves."
6. "Why do banks and telecom companies depend so heavily on Oracle?"
7. "What makes Oracle's architecture fundamentally different from MySQL or PostgreSQL?"

□ For Part 3 — Reasoning

1. "Why would a bank choose Oracle over MySQL? Frame the answer in terms of risk, regulation, and downtime cost."
2. "Why would a startup choose PostgreSQL over Oracle? Frame the answer in terms of cost, agility, and cloud-native fit."
3. "Why do some enterprises remain loyal to Microsoft SQL Server even when alternatives are free?"
4. "How has the rise of cloud infrastructure changed how companies pick a database in 2024–2026?"
5. "Why do enterprises pay millions for commercial databases when PostgreSQL is free?"

□ For Part 4 — Architecture

1. "Build a comparison matrix of Oracle, SQL Server, PostgreSQL, and MySQL across 8 architectural dimensions."
2. "Which DBMS is most commonly used in: enterprise systems, startups, government, cloud-native, banking, large transactional environments? Justify each."
3. "Compare the high-availability options of all four DBMSs."

4. "Compare backup and recovery strategies across the four DBMSs."

For Part 5 — Migration

1. "If all DBMSs support tables, joins, transactions, and procedures — why do companies still use different ones? Give 7 reasons."
2. "Give 3 real-world examples of major database migrations and explain why they happened."
3. "List the technical challenges that appear during a database migration from Oracle to PostgreSQL."
4. "Why is database migration one of the highest-risk projects in enterprise IT?"
5. "How does a database architecture decision made today affect a company for the next 10–20 years?"

PART C — HOW WE COMBINED GOOGLE + AI

The real power was using **both together**. Here's the exact loop we followed for every topic:

- ```

1. Ask AI for an overview
 ↓
2. Google the specific facts AI mentioned
 ↓
3. Read official documentation to confirm
 ↓
4. Ask AI to compare or simplify
 ↓
5. Search for real-world case studies
 ↓
6. Ask AI "devil's advocate" questions
 ↓
7. Combine everything in our own words

```

## 7. A Full Example — How We Researched One Topic

To make this concrete, here is how we actually researched **"Why do banks use Oracle instead of MySQL?"** step by step.

## Step 1 — Asked AI for an overview

Prompt: "Why do banks prefer Oracle over MySQL? Give technical, business, and regulatory reasons."

AI mentioned: RAC, Data Guard, ACID guarantees, compliance, vendor support.

## Step 2 — Googled the specific facts

Search: `oracle RAC site:oracle.com` → Confirmed RAC is Enterprise Edition only, used by major banks. Search: `oracle data guard high availability` → Confirmed real-time standby capability.

## Step 3 — Read official docs

Visited: `docs.oracle.com/en/database/oracle/oracle-database/19/haovw/` to confirm the high-availability architecture.

## Step 4 — Asked AI to simplify

Prompt: "Explain Oracle Data Guard in one paragraph for a beginner."

## Step 5 — Searched for real case studies

Search: `"core banking system" oracle case study` → Found references to HSBC, Citi, and Flexcube/Temenos T24 being Oracle-based.

## Step 6 — Asked AI for the counter-argument

Prompt: "Now argue the opposite — why might a modern bank move some workloads to PostgreSQL or Aurora?"

## Step 7 — Wrote our final paragraph

Combined every source + our own reflection:

"A bank deals with billions of dollars of transactions, strict regulations, and zero tolerance for data loss. Oracle provides mature high-availability features (RAC, Data Guard), advanced security, and enterprise-grade 24/7 vendor support. For a bank, the high license cost of Oracle is insurance against catastrophic risk — far cheaper than a data breach or hours of downtime."

This is **much better** than what any single search or single AI prompt would have given us.

# PART D — VERIFICATION & QUALITY CONTROL

## 8. The "Rule of 2 Sources"

Any fact, number, or claim that went into the final report had to be confirmed by **at least 2 trustworthy sources**.

| Example claim                                   | Verified with                                             |
|-------------------------------------------------|-----------------------------------------------------------|
| "Oracle XE has a 12 GB user data limit"         | Oracle official XE docs + Oracle blog post                |
| "Amazon migrated off Oracle in 2019"            | AWS official blog + multiple news articles                |
| "PostgreSQL is the most popular open-source DB" | DB-Engines ranking + Stack Overflow Developer Survey 2024 |
| "Oracle RAC is Enterprise Edition only"         | Oracle licensing PDF + Oracle documentation               |

## 9. How We Avoided Common AI Mistakes

| Risk                  | How we avoided it                                                   |
|-----------------------|---------------------------------------------------------------------|
| Hallucinated facts    | Verified every number and statistic with official docs              |
| Outdated info         | Asked AI to focus on "2024-2026" and confirmed with recent articles |
| Generic answers       | Used role-based and structured prompts to force depth               |
| Copy-paste plagiarism | Always rewrote AI answers in our own words                          |
| One-sided opinions    | Used devil's advocate prompts to get balanced views                 |

## 10. How We Avoided Common Google Mistakes

| Risk                          | How we avoided it                                                   |
|-------------------------------|---------------------------------------------------------------------|
| <b>Sponsored / ad content</b> | Skipped paid results at the top of the page                         |
| <b>SEO blog spam</b>          | Stuck to Tier 1 and Tier 2 sources                                  |
| <b>Outdated articles</b>      | Filtered with <code>2024..2026</code> and checked publication dates |
| <b>Single-vendor bias</b>     | Read articles from competing vendors                                |
| <b>Surface-level content</b>  | Always clicked through to the official documentation                |

# PART E — ORGANIZING THE INFORMATION □

## 11. Our Note-Taking Template

For every concept we researched, we filled this template:

```
Topic: [e.g., Oracle Multitenant Architecture]

What I learned (in my own words):
- ...

Key facts (with source):
- Fact 1 — Source: https://docs.oracle.com/...
- Fact 2 — Source: https://db-engines.com/...

Real-world example:
- ...

My reflection / opinion:
- ...

Questions still unclear:
- ...
```

## 12. How We Structured the Final Document

After collecting verified notes from Google + AI, we built the final document in this order:

1. Grouped notes by task part (Part 1, 2, 3, 4, 5)

2. Built tables for comparisons (clearer than prose)
3. Rewrote everything in professional engineering tone
4. Removed AI-style filler phrases ("It's important to note that...")
5. Added personal reflection in Parts 3 and 5
6. Added a References list with every verified source

## PART F — TOOLS WE USED □

| Tool                          | Purpose                                |
|-------------------------------|----------------------------------------|
| Google Search                 | Primary research engine                |
| Google Scholar                | Academic / architectural papers        |
| DB-Engines.com                | DBMS popularity ranking                |
| YouTube                       | Visual explanations of architecture    |
| Stack Overflow                | Real engineer opinions                 |
| Reddit (r/database, r/oracle) | Community discussions                  |
| AI Assistant                  | Brainstorming, structuring, comparing  |
| Markdown editor               | Note-taking and final document writing |
| Draw.io                       | Architecture diagrams (CDB/PDB)        |
| Grammarly                     | Polishing the final document           |

## PART G — LESSONS LEARNED □

After completing the full investigation, we identified these key lessons:

1. **Google is for finding facts. AI is for explaining and structuring them.** Use both.
2. **The quality of the search/prompt determines the quality of the answer.** Specific = good. Vague = useless.
3. **Always verify with official documentation.** Random blogs and AI can both be wrong.
4. **AI is great for comparisons** because it can hold many concepts in mind at once.
5. **Real case studies (Amazon, Uber, HSBC) are more powerful than abstract explanations.**

6. **Devil's advocate questions** produce the most balanced answers.
7. **Rewriting in your own words** is what actually turns information into understanding.
8. **Speed-read articles using the Skim → Spot → Capture method** — never read every word.

## 13. Conclusion — A Modern Research Methodology

The way we researched this task reflects how **modern engineers do research in real workplaces**:

- □ **Search smart on Google** — keywords, operators, trusted sources
- □ **Verify with official documentation** — never trust a single blog
- □ **Use AI to brainstorm, compare, and structure** — but never as the final source
- □ **Cross-check everything** with the Rule of 2 Sources
- ✍️ **Rewrite in your own voice** to truly understand the material

This combined methodology — **Google + Documentation + AI + Verification + Reflection** — is what turned this task from a copy-paste exercise into a genuine learning experience and a professional engineering reflection document.

### □ Summary of Files Produced

| File                          | Purpose                                         |
|-------------------------------|-------------------------------------------------|
| 01_Task_Explanation.md        | Explains the assignment in simple language      |
| 02_How_To_Research_Guide.md   | General reusable research manual                |
| FINAL_Reflection_Document.md  | The complete, ready-to-submit reflection report |
| HOW_We_Used_AI_To_Research.md | This file — Google + AI research methodology    |

End of research methodology document.